

dition, a large number of PCBs are being imported as fully-assembled boards. These go straight into finished products assembled in India as part of the low-value-added manufacturing model followed by most companies in India.

Government policies and their impact

The industry feels that the Indian government does not take a proactive approach to encourage PCB manufacturing in India. The current government policies benefit only the large industries. The bureaucratic hurdles are too much for the small-scale industry to handle. The processes to get approvals and clearances for projects are lengthy, complicated and consume a lot of time, energy and money.

The Central and state governments can play a major role in establishing PCB industry in India, which is a high-value-added manufacturing activity requiring high capital investment. The industry needs government support for access to technology, low-cost finance and infrastructure to establish PCB manufacturing clusters where the entire value chain can flourish in a supportive environment.

According to Amit Agrawal, director, Advance Technologies, corruption is a major curse on our system. He mentions that two Indian firms, that had started manufacturing PCB laminates, closed down recently due to the poor policies and 'torture' from the government.

Moreover, duty-free import for PCBs has encouraged dumping of cheap, Chinese PCBs in India, reducing the demand for Indian PCBs. But some policies introduced by the government of India in last two years hold a ray of hope for the Indian PCB industry.

Modified Special Incentive Package Scheme (M-SIPS). M-SIPS is an investment-based scheme that is aimed to offset disability and attract investment in almost all verticals of electronic system design and manufacturing (ESDM) sector. According to Rajneesh Garg,

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managing director, Garg Electronics, "This would be of value only to large industries." However, some say that there are provisions for smaller industries, and that it is a wrong perception that M-SIPS can be of value to only large industries.

Preferential Market Access (PMA).

The revised PMA policy mandates local sourcing of electronics for government departments, except the Ministry of Defence, for a period of 10 years. While the revised policy mandates sourcing of sensitive telecom and electronic equipment (and technology) from domestic equipment makers, it has exempted the private sector from the changed rules. However, it will cover the schemes and grants made by the Central government. This will give impetus to the demand of PCBs in India and eventually boost the indigenous manufacturing in the country.

Looking towards the future

India has a good number of PCB manufacturers in the country. However, the local capability is restricted to producing single-, dual- and multi-layered PCBs of up to 12 layers, and to 16 to 24 layered PCBs in some cases. The PCBs needed in products with high demand, like telecom, consumer and automotive applications, require complex multi-layered PCBs that are not available locally, thus necessitating imports.

Building on the existing capability, special initiatives need to be offered to encourage indigenous design and manufacturing of multi-layered PCBs of up to 64 layers and beyond. This

Some solutions that the government can implement to support this industry are:

- Elimination of inverted duty structure
- Sorting out excise and customs duty procedural issues
- High import tariff on PCB imports. Nil duty on raw material imports.
- Imposing anti-dumping duty or safeguard duty
- Interest subsidies or low-interest loans for PCB manufacturers
- Quicker, easier reimbursement of CENVAT

will ensure local sourcing of PCBs by product manufacturers, thus contributing to higher value addition. This shall also meet the requirements of the local EMS players who source PCBs in bulk. This would go well with DeitY strategy of PMA, which enlists PCB and its contribution to bill of materials in its notifications.

India possesses a good chance of elevating itself as a global supplier of PCBs. The country is already exporting 20 to 30 per cent of the total production of PCBs. However, the volumes and the subsequent price points in comparison with China is a major hindrance as is the ability of Indian PCB makers to handle complex products that require more than 24 layers. Government support in the form of lower excise duties and rebates for up-gradation of capital equipment could make a positive impact. ●

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